Fiscal Year:	FY 2012	**Task Last Updated:**	FY 10/12/2011
PI Name:	Mulavara, Ajitkumar P. Ph.D.		
Project Title:	Development of Countermeasures to Aid Functional Egress from the Crew Exploration Vehicle Following Long Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Formerly at Universities Space Research Association		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	10/01/2009	**End Date:**	09/30/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bloomberg, Jacob (NASA Johnson Space Center) Cohen, Helen (Baylor College of Medicine) Reschke, Millard (NASA Johnson Space Center) Wood, Scott (NASA Johnson Space Center) Peters, Brian (Wyle Laboratories)		
Grant/Contract No.:	NCC 9-58-SA02001		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Crewmember adapted to the microgravity state may need to egress the vehicle within a few minutes for safety and operational reasons in various sea state conditions following a water landing. The act of emergency egress includes and is not limited to rapid motor control tasks (including both fine motor such as object manipulation and gross motor such as opening a hatch) and visual acuity tasks while maintaining spatial orientation and postural stability. Exposure to even low frequency motions (0.2-2.0 Hz) induced by sea conditions surrounding a vessel can cause significant fine and gross motor control problems affecting critical functions. These motion frequencies coupled with the varying sea state conditions (sea states 1-7 indicate frequencies ranging from 0.125-0.5 Hz, CxP 70023) can thus cause performance deficits by affecting the efficacy of motor and visual acuity dependent skills in tasks critical to emergency egress activities. Further, astronauts experience disturbances in sensorimotor function following their return to Earth due to adaptive responses that occur during exposure to the microgravity conditions of spaceflight. Thus, during exploration class missions the interactions between a debilitated crewmember during re-adaptation to Earth's gravity and the environmental constraints imposed by a water landing scenario may lead to disruption in the ability to perform functional egress tasks during the initial introduction to the Earth's gravitational environment. At present, no operational countermeasure has been implemented to mitigate this risk. Therefore, the overall goals of this project are to: 1) Investigate performance of motor and visual tasks during simulated varying sea state conditions with perturbations in the frequency range from 0.2-2.0 Hz. 2) Develop a countermeasure based on stochastic resonance that could be implemented to enhance sensorimotor capabilities with the aim of facilitating rapid adaptation to gravitational transitions following long-duration spaceflight. The objective of two separate studies that were conducted was to document human visual performance during simulated wave motion in the 0.1 to 2.0 Hz range. In the first study we examined, in 12 healthy subjects, the changes in accuracy when performing a seated visual target acquisition task in which the location of target was offset vertically during horizontal full-body rotation at an oscillating frequency of 0.8 Hz (peak velocity of 160 deg/s). The main finding was that the accuracy of performance degraded in 7 of 12 subjects when acquiring vertical targets at perturbing frequencies of 0.8 Hz in the horizontal plane by one step size. In the second study we examined 112 normals and 45 patients who had been diagnosed with either unilateral vestibular weaknesses or with post-acoustic neuroma resections. Subjects sat in a chair that could oscillate vertically with the head either free or constrained with a cervical orthosis. We determined their seated dynamic visual acuity (DVA) task performance while the chair was stationary and during vertical full-body oscillations at perturbing frequencies of 2 Hz (peak-to-peak motion of 5 cm). Scores were worse for both groups during the dynamic condition compared to the static condition. In the dynamic condition patients's scores were significantly worse than normals's scores. Dynamic visual acuity (DVA) is reduced in the vertical plane at frequencies of 2 Hz and in the horizontal plane at frequencies of 0.8 Hz. DVA varies with target location, with acuity optimized for targets in the plane of motion. Thus, low frequency perturbations in the horizontal and vertical planes can cause decrements in visual performance. Perturbations at low frequency motions (0.1-2 Hz) may exacerbate sensorimotor deficits after spaceflight. We have completed a study on the development of a countermeasure with the aim of facilitating rapid adaptation to gravitational transitions following long-duration spaceflight. Stochastic resonance (SR) is a mechanism by which noise can assist and enhance the response of neural systems to relevant sensory signals. Application of imperceptible SR noise coupled with sensory input through the proprioceptive, visual, or vestibular sensory system has been shown to improve motor function. Specifically, studies have shown that vestibular electrical stimulation by imperceptible stochastic noise (SRVS), when applied to normal young and elderly subjects, significantly improved their ocular stabilization reflexes in response to whole-body tilt as well as balance performance during postural disturbances. Our study showed that low imperceptible levels of white noise based electrical stimulation of the vestibular system improves balance performance consistent with the stochastic resonance phenomenon in normal healthy control subjects. The amplitude of optimal stimulus for improving balance performance was predominantly in the range of 100-400 μA. An SRVS based device may be fielded, either as a training modality to enhance adaptability or skill acquisition, or as a miniature patch type stimulator that may be worn by astronauts to enhance adaptation following gravitational transitions including people with disabilities due to aging or disease, improving posture and locomotion function.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Ultimately an SR based countermeasure might be fielded either as a training modality to enhance adaptability or skill acquisition, or as a miniature patch type stimulator that may be worn by people with disabilities due to aging or disease improving posture and locomotion function. Such populations may include older adults; patients with either diabetic neuropathy or stroke.
Task Progress:	1) Visual acuity was measured in 112 normal subjects and 45 subjects with vestibular impairments. Subjects sat in a comfortable chair while viewing a LCD screen at 2 m. The size of optotypes presented ranged from 20/8 to 20/200 Snellen ratios, in one of 8 configurations for 75 ms around peak velocity of motion. In the dynamic testing phase, the chair moved vertically, at 2 Hz, ± 5 cm to mimic vertical oscillations during walking. The Best PEST psychophysical threshold detection algorithm was used to determine the visual acuity threshold for each condition. Results indicated that visual acuity is significantly reduced during 2 Hz vertical translational motion compared to static, across all age ranges within and between control and clinical vestibular subjects. 2) Visual acuity was measured in 12 healthy subjects. Acuity thresholds were established using Landolt C optotypes presented on a LCD screen at 1.3 m. In dynamic tests chair oscillated at 0.8 Hz at peak velocity 60 deg/sec. All acuity measures were made using a forced-choice strategy. To measure the effect of target location optotypes were randomly presented at nine different locations on the screen offset upto 10 degrees. The optotype size was set to Snellen ratios of 20/20, 20/30 or 20/50. Presentation duration was set to 150, 300 and 450 ms. All these conditions were counterbalanced across 5 trials. Dynamic visual acuity threshold was reduced relative to static acuity in 7 of 12 subjects by one step size during horizontal rotational motion at 0.8 Hz. Both accuracy and reaction time varied as a function of target location, with greater performance decrements when acquiring vertical targets. 3) 15 subjects performed a standard balance task of standing on a block of foam with their eyes closed. Bipolar stochastic electrical stimulation was applied to the vestibular system using a constant current stimulator through electrodes placed over the mastoid process behind the ears. Amplitude of the signals varied in the range of 0-700 microamperes. Balance performance was measured using a force plate and inertial motion sensors placed on the trunk

	and head. Balance performance with stimulation was significantly improved in the range of 10%-25% compared with no stimulation. The signal amplitude at which performance was maximized was in the range of 100-400 microamperes. In the current year we are implementing studies to determine the efficacy of vestibular stochastic resonance in 1) physiological and perceptual responses under canal-otolith conflict perturbations using a Variable radius centrifuge 2) balance and locomotor performance on unstable platforms. Further we are investigating if we can predict the efficacy of the countermeasure using tests of sensory-preferences. This project has resulted in 7 Abstracts presented at 4 different meetings, 2 Invention Disclosures, and 1 article in peer reviewed journals.
Bibliography Type:	Description: (Last Updated: 08/25/2020)
Abstracts for Journals and Proceedings	Appelbaum ML, De Dios YE, Kulecz WB, Peters BT, Wood SJ. "Effect of target location on dynamic visual acuity during passive horizontal rotation." Presented at The 34th MidWinter Meeting of the Association for Research in Otolaryngology, Baltimore, MD, February 19-23, 2011. 34th MidWinter Meeting of the Association for Research in Otolaryngology, Baltimore, MD, February 19-23, 2011. , Feb-2011
Abstracts for Journals and Proceedings	Fiedler MJ, Cohen H, Mulavara AP, Peters B, Miller C, Bloomberg JJ. "Metrics of balance control for use in screening tests of vestibular function." 8th Symposium on the Role of the Vestibular Organs in Space, Houston, TX, April 8-10, 2011. 8th Symposium on the Role of the Vestibular Organs in Space, Houston, TX, April 8-10, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Mulavara A, Fiedler M, Kofman I, Wood S, Serrador J, Peters B, Cohen H, Reschke M, Bloomberg J. "Vestibular stochastic resonance as a method to improve balance function: Optimization of stimulus characteristics." Presented at Neuroscience 2010, Society for Neuroscience's 40th Annual Meeting, San Diego, CA, November 13-17, 2010. Neuroscience 2010, Society for Neuroscience's 40th Annual Meeting, San Diego, CA, November 13-17, 2010. , Nov-2010
Abstracts for Journals and Proceedings	Mulavara AP, Fiedler M, Heap E, Acock K, De Dios Y, Kofman I, Peters B, Wood S, Serrador J, Cohen H, Reschke M, Bloomberg J. "Optimization of stimulus characteristics for vestibular stochastic resonance to improve balance function." The 34th Mid Winter Meeting of the Association for Research in Otolaryngology (ARO), Baltimore, MD, February 19-23, 2011. 34th MidWinter Meeting of the Association for Research in Otolaryngology, Baltimore, MD, February 19-23, 2011. , Feb-2011
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Abstracts for Journals and Proceedings	Mulavara AP, Peters B, Wood SJ, Cohen H, Kulecz WB, Miller C, Reschke MF, Bloomberg JJ. "Effect of passive horizontal rotations and vertical oscillations on dynamic visual acuity." Presented at the 8th Symposium on the Role of the Vestibular Organs in Space, Houston, TX, April 8-10, 2011. 8th Symposium on the Role of the Vestibular Organs in Space, Houston, TX, April 8-10, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Mulavara AP, Wood SJ, Fiedler MJ, Kofman I, Kulecz WB, Miller C, Peters B, Serrador JM, Cohen H, Reschke MF, Bloomberg JJ. "Visual performance challenges to low-frequency perturbations after long-duration space flight and countermeasure development." Presented at the 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Articles in Peer-reviewed Journals	Mulavara AP, Fiedler MJ, Kofman IS, Wood SJ, Serrador JM, Peters B, Cohen HS, Reschke MF, Bloomberg JJ. "Improving balance function using vestibular stochastic resonance: optimizing stimulus characteristics." Experimental Brain Research, 2011 Apr;210(2):303-12. Epub 2011 Mar 26. PubMed PMID: 21442221 ; http://dx.doi.org/10.1007/s00221-011-2633-z , Apr-2011
Awards	Mulavara A. "Scientist of the Month, Division of Space Life Sciences, Universities Space Research Association (USRA), August 2011." Aug-2011